

INFLATION CONTROL, COMPLIANCE ANALYSIS, AND LEAK DETECTION EMBODIMENTS

In various embodiments, the intraluminal compliance and distance profiling device incorporates controlled inflation systems configured to regulate expansion of inflatable regions in a repeatable and safe manner while simultaneously extracting physiologic compliance information from measured pressure responses.

Inflation control may be implemented using manual pumps, electronically actuated pumps, micro-compressors, syringe mechanisms, or fluid displacement systems. Control logic may regulate inflation rate, maximum pressure thresholds, staged expansion steps, or timed inflation cycles designed to minimize discomfort while improving measurement consistency.

In certain embodiments, closed-loop inflation control is performed using real-time pressure feedback obtained from integrated pressure sensors, strain sensors, or deformation sensing elements. Feedback control algorithms may include proportional, integral, derivative (PID) control, threshold-based regulation, adaptive control routines, or model-based control strategies.

Measurement workflows may include staged inflation sequences in which the inflatable cuff expands incrementally, pauses to allow stabilization, records compliance measurements, and proceeds to subsequent expansion stages. Such staged sequences improve repeatability and reduce motion artifacts.

The system may derive compliance characteristics by analyzing pressure-versus-expansion relationships, inflation rate responses, hysteresis behavior during inflation and deflation, or time-dependent relaxation characteristics of surrounding tissue structures. Derived parameters may include compliance indices, stiffness metrics, deformation gradients, or stability coefficients.

In certain implementations, leak detection algorithms monitor pressure decay during hold phases. Unexpected pressure loss may indicate incomplete sealing, improper positioning, hardware leakage, or user movement. The device may prompt corrective action or automatically terminate measurement if leak thresholds are exceeded.

Inflation safety mechanisms may include software-enforced pressure ceilings, expansion rate limits, emergency deflation triggers, mechanical relief valves, or automatic pump shutdown conditions operating redundantly through hardware and software pathways.

The device may further perform calibration routines using known internal volumes or reference expansion behaviors to verify inflation system performance prior to measurement sessions. Calibration checks may occur during docking, initialization, or scheduled maintenance intervals.

In certain embodiments, inflation control parameters adapt dynamically based on prior measurements, selected anatomy mode, or detected compliance characteristics, thereby personalizing measurement protocols to individual physiologic response patterns.

Accordingly, the disclosed inflation control and compliance analysis embodiments enable safe, repeatable expansion control while extracting quantitative mechanical information, transforming the inflatable region into an active sensing mechanism within the broader physiologic measurement system.